Flory exponent for a polymer globule

Complex disordered systems: Polymers

Consider a single flexible polymer of N monomers, each of size b, in a poor solvent (i.e., $T < \Theta$), so that the polymer collapses into a dense globule.

1. The entropic contribution for the chain can be approximated (see notes) with

$$S = \frac{-3k_B R^2}{2Nb^2}$$

where R is the radius of the globule (essentially proportional to the root mean squared end-to-end distance).

Try to figure out why this leads to what is called the "elastic" free energy of the chain.

2. The globule is dense. Write the interaction free energy using the virial expansion, noting that $\rho = N/R^3$ stopping at order three.

We state that the second virial drives the collapse (attraction) while the third virial is repulsive and prevents complete collapse.

3. Minimise the total free energy to obtain a function $f(R) = 0$. Show that if we assume that R grows with N more slowly than in the theta conditions, the entropic part of the resulting function of R can be neglected.

4. Proceed with the approximation and extract the Flory exponent ν for the globule, defined by $R \sim N^\nu$: how is it different for other solvent conditions?

Solution

1. For the entropy

$$S = \frac{-3k_B R^2}{2Nb^2}$$

then the entropic free energy contribution is

$$F_{\text{ent}} = -TS = \frac{3k_B T R^2}{2Nb^2}$$

which is quadratic in R like an elastic energy.

2. We can write

$$\frac{F_{\text{int}}}{k_B T} \sim R^3 \left(B_2 \rho^2 + B_3 \rho^3\right)$$

or

$$F_{\text{int}}(R) \sim k_B T \left(\frac{vN^2}{R^3} + \frac{wN^3}{R^6}\right).$$

where we renamed the second virial coefficient $B_2 = v$ and the third virial coefficient $B_3 = w$. In a poor solvent $v < 0$.

3. Sum the two entropic and interacting contributions:

$$F(R) \sim k_B T \left[\frac{3R^2}{2Nb^2} + \frac{vN^2}{R^3} + \frac{wN^3}{R^6} \right]$$

At equilibrium, the free energy is minimised with respect to R.
Set $\frac{dF}{dR} = 0$:

$$\frac{dF}{dR} = k_B T \left(\frac{3R}{Nb^2} - 3vN^2 R^{-4} - 6wN^3 R^{-7} \right) = 0$$

Multiply by NR^7

$$\frac{3R^8}{b^2} - 3vN^3 R^3 - 6wN^4 = 0$$

If $R \sim N^\nu$ with $\nu < 1/2$ (i.e., the chain is more collapsed than in theta conditions) then the first term scales as $N^{8\nu}$, the second as $N^{3+3\nu}$ and the third as N^4.
For $\nu = 1/2$ the first term scales as N^4, the second as $N^{4.5}$ and the third as N^4.
For smaller ν the first term becomes subleading with respect to the other two.

4.

So we approximate the minimisation condition as

$$-3vN^3 R^3 - 6wN^4 = 0$$

and, since $v < 0$, get

$$R^3 = \frac{2wN}{|v|} \quad \Rightarrow \quad R \propto N^{1/3}$$

So the Flory exponent of a globule is $\nu = 1/3$. Since $R^3 = 2wN/|v|$, the globule has a fixed density $\rho = N/R^3 \simeq |v|/2w$, independent of N: its volume simply grows linearly with the number of monomers, as for a dense sphere (something you could have guessed without the calculation).
This answers the last question: as the solvent gets worse, ν decreases, from $\nu \approx 3/5$ in a good solvent (swollen coil), to $\nu = 1/2$ at the θ-point (ideal chain), to $\nu = 1/3$ in a poor solvent (compact globule).